



ASSIGNMENT 1

U26AI103
NIHIT BHAGWAT
AI BTECH
B-33



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Introduction to programming [AI103] Lab

1. PROGRAM TO CALCULATE SIMPLE INTEREST

```
1  #include <stdio.h>
2  int main()
3  {
4      float principle, rate, time, intrest;
5      printf("Enter the principle amount: \n");
6      scanf("%f", &principle);
7      printf("Enter the amount of interest: \n");
8      scanf("%f", &rate);
9      printf("Enter the time period: \n");
10     scanf("%f", &time);
11     intrest = (principle * rate * time) / 100;
12     printf("The simple interest is: %f\n", intrest);
13     return 0;
14 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc sim
Enter the principle amount:
25000
Enter the amount of interest:
12
Enter the time period:
6
The simple interest is: 18000.000000
PS C:\VS CODE\cs_assignment_1> █
```

2. PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL AND PERCENTAGE.

```
1  #include <stdio.h>
2  int main()
3  {
4      float s1, s2, s3, s4, s5, total, percentage;
5      printf("Enter marks of 1st subject : \n");
6      scanf("%f", &s1);
7      printf("Enter marks of 2nd subject : \n");
8      scanf("%f", &s2);
9      printf("Enter marks of 3rd subject : \n");
10     scanf("%f", &s3);
11     printf("Enter marks of 4th subject : \n");
12     scanf("%f", &s4);
13     printf("Enter marks of 5th subject : \n");
14     scanf("%f", &s5);
15     printf("\n\n");
16     if(s1<=100 && s1>=0){
17         printf("s1 : %.2f\n", s1);
18     }
19     else{
20         printf("s1 : Not valid marks");
21     }
22     if(s2<=100 && s2>=0){
23         printf("s2 : %.2f\n", s2);
24     }
25     else{
26         printf("s2 : Not valid marks");
27     }
28     if(s3<=100 && s3>=0){
29         printf("s3 : %.2f\n", s3);
30     }
31     else{
32         printf("s3 : Not valid marks");
33     }
34     if(s4<=100 && s4>=0){
35         printf("s4 : %.2f\n", s4);
36     }
37     else{
38         printf("s4 : Not valid marks");
39     }
40     if(s5<=100 && s5>=0){
41         printf("s5 : %.2f\n", s5);
42     }
43     else{
44         printf("s5 : Not valid marks\n");
45     }
46     total=s1+s2+s3+s4+s5;
47     percentage=(total/5);
48     if((s1>=0 && s1<=100) && (s2>=0 && s2<=100) && (s3>=0 && s3<=100) && (s4>=0 && s4<=100) && (s5>=0 && s5<=100) ){
49         printf("Total marks is : %.2f\n", total);
50         printf("Total percentage is : %.2f\n", percentage);
51     }
52     else{
53         printf("Total marks : ERROR\n");
54         printf("Total percentage : ERROR\n");
55     }
56     return 0;
57 }
```

OUTPUT:

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\lab_1"
Enter marks of 1st subject :
96
Enter marks of 2nd subject :
98
Enter marks of 3rd subject :
95
Enter marks of 4th subject :
63
Enter marks of 5th subject :
91

s1 : 96.00
s2 : 98.00
s3 : 95.00
s4 : 63.00
s5 : 91.00
Total marks is : 443.00
Total percentage is : 88.60
PS C:\VS CODE\cs_assignment_1\lab_1> █
```

3. PROGRAM TO CALCULATE GROSS SALARY.

```
1  #include<stdio.h>
2  int main()
3  {
4      float salary, CTC, employee_benefits;
5      printf("Enter your CTC salary: \n");
6      scanf("%f", &CTC);
7      printf("Enter your employee benefits: \n");
8      scanf("%f", &employee_benefits);
9      salary = CTC - employee_benefits;
10     printf("your gross salary is: %f", salary);
11 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc pr
e_gross_salary }
Enter your CTC salary:
120000
Enter your employee benefits:
20000
your gross salary is: 100000.000000
PS C:\VS CODE\cs_assignment_1> 
```

4. PROGRAM TO CONVERT TEMPRATURE FROM FAHRENHEIT TO CENTIGRADE DEGREES.

```
1  #include<stdio.h>
2  int main()
3  {
4      float fahrenheit, celsius;
5      printf("Enter the temprature in fahrenheit: \n");
6      scanf("%f", &fahrenheit);
7      celsius = (fahrenheit - 32)*5/9;
8      printf("The temprature in celsius is = %f", celsius);
9      return 0;
10 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc fahrenheit.c
Enter the temprature in fahrenheit:
110
The temprature in celsius is = 43.333332
PS C:\VS CODE\cs_assignment_1> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc fahrenheit.c
Enter the temprature in fahrenheit:
120
The temprature in celsius is = 48.888889
PS C:\VS CODE\cs_assignment_1> 
```

5. PROGRAM TO SWAP TWO VARIABLES USING THIRD NUMBER.

```
1  #include <stdio.h>
2  int main()
3  {
4      int first, second, third;
5      printf("Enter two numbers : ");
6      scanf("%d %d", &first, &second);
7      //now swap two numbers by third//
8      third = first;
9      first=second;
10     second=third;
11     printf("After swappping the numbers we get : %d %d", first, second);
12 }
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc swapping_by_third_number.
Enter two numbers : 5
6
After swappping the numbers we get : 6 5
PS C:\VS CODE\cs_assignment_1> |
```

6. PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLES.

```
1  #include <stdio.h>
2  int main()
3  {
4      int a, b;
5      printf("Enter two numbers: \n");
6      scanf("%d %d", &a, &b);
7      a=a+b;
8      b=a-b;
9      a=(a-b);
10     printf("After swapping the numbers : %d %d", a, b);
11     return 0;
12 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc swappin
er }
Enter two numbers:
9
19
After swapping the numbers : 19 9
PS C:\VS CODE\cs_assignment_1> 
```

7. PROGRAM TO CALCULATE AREA OF TRIANGLE

```
cs_assignment_1 > area_of_triangle.c > main()
1  #include<stdio.h>
2  #include<math.h>
3  int main()
4  {
5      float a,b,c,s,area;
6      printf("Enter three sides of a triangle: \n");
7      scanf("%f %f %f", &a, &b, &c);
8      s=(a+b+c)/2;
9      area=sqrt(s*(s-a)*(s-b)*(s-c));
10     printf("Area of the triangle is = %f", area);
11     return 0;
12 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc a
Enter three sides of a triangle:
5
6
7
Area of the triangle is = 14.696939
PS C:\VS CODE\cs_assignment_1> 
```

8. EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS. WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAY, HOURS AND MINUTES

```
cs_assignment_1 / ... earth.c ...
1  #include<stdio.h>
2  int main()
3  {
4      float seconds, days, hours, minutes;
5      seconds = 31558150;
6      minutes = seconds/60;
7      hours = minutes/60;
8      days = hours/24;
9      printf("Total number of days in a year is: %f\n", days);
10     printf("total no. of hours in a years is: %f\n", hours);
11     printf("total no. of minutes in a year is : %f\n", minutes);
12     return 0;
13 }
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1" ; if ($?) { gcc earth.c -o earth } ; if ($?) { .\earth }
Total number of days in a year is: 365.256378
total no. of hours in a years is: 8766.153320
total no. of minutes in a year is : 525969.187500
PS C:\VS CODE\cs_assignment_1> |
```

9. PROGRAM TO READ TIME IN HR, MIN, SEC AND CONVERT IT INTO TOTAL SECOND.

```
1  #include <stdio.h>
2  int main()
3  {
4      float time, hours, minutes, seconds, total_seconds;
5      printf("Enter hours:\n");
6      scanf("%f", &hours);
7      printf("Enter minutes:\n");
8      scanf("%f", &minutes);
9      printf("Enter seconds:\n");
10     scanf("%f", &seconds);
11     total_seconds = ((hours*60)*60)+(minutes*60)+seconds;
12     printf("Total time in seconds is : %f", total_seconds);
13     return 0;
14 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 9_time.c -o 9_
Enter hours:
8
Enter minutes:
35
Enter seconds:
53
Total time in seconds is : 30953.000000
PS C:\VS CODE\cs_assignment_1> |
```

10. WRITE A C PROGRAM TO CALCULATE THE CUT OFF MARK OF A STUDENT USING THE FORMULA.

$$CM = M/2 + P/2 + C/2 + E$$

WHERE CM = Cut off mark

M = Marks in Mathematics out of 200

P = Marks in Physics out of 200

C = Marks in Chemistry out of 200

E = Marks in entrance examination out of 100.

```
1  #include <stdio.h>
2  int main()
3  {
4      float M, P, C, E, total_marks, CM;
5      printf("Enter marks of maths :\n");
6      scanf("%f", &M);
7      printf("Enter marks of physics :\n");
8      scanf("%f", &P);
9      printf("Enter marks of chemistry :\n");
10     scanf("%f", &C);
11     printf("Enter marks of Entrance exam :\n");
12     scanf("%f", &E);
13     if(M<=200)
14         printf("Marks obtained in Maths is : %f\n", M);
15     else
16         printf("Not valid\n");
17     if(P<=200)
18         printf("Marks obtained in Physics is : %f\n", P);
19     else
20         printf("Not valid\n");
21     if(C<=200)
22         printf("Marks obtained in Chemistry is : %f\n", C);
23     else
24         printf("Not valid\n");
25     if(E<=100)
26         printf("Marks obtained in Entrance exam is : %f\n", E);
27     else
28         printf("Not valid\n");
29     CM = M/2 + P/2 + C/2 + E;
30     if(M<=200 && P<=200 && C<=200 && E<=100)
31         printf("Cut off marks of student is : %f", CM);
32     else
33         printf("CUT OFF MARKS : INVALID\n");
34     return 0;
35 }
```

OUTPUT:

```
1  #include <stdio.h>
2  int main()
3  {
4      float M, P, C, E, total_marks, CM;
5      printf("Enter marks of maths :\n");
6      scanf("%f", &M);
7      printf("Enter marks of physics :\n");
8      scanf("%f", &P);
9      printf("Enter marks of chemistry :\n");
10     scanf("%f", &C);
11     printf("Enter marks of Entrance exam :\n");
12     scanf("%f", &E);
13     if(M<=200)
14         printf("Marks obtained in Maths is : %f\n", M);
15     else
16         printf("Not valid\n");
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 10_
Enter marks of maths :
196
Enter marks of physics :
189
Enter marks of chemistry :
173
Enter marks of Entrance exam :
86
Marks obtained in Maths is : 196.000000
Marks obtained in Physics is : 189.000000
Marks obtained in Chemistry is : 173.000000
Marks obtained in Entrance exam is : 86.000000
Cut off marks of student is : 365.000000
PS C:\VS CODE\cs_assignment_1> |
```

11. PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME.

```
1  #include<stdio.h>
2  int main()
3  {
4      int total_seconds, hours, minutes, seconds;
5      printf("Enter total seconds: \n");
6      scanf("%d", &total_seconds);
7      hours = total_seconds/3600;
8      minutes = total_seconds%3600/60;
9      seconds = total_seconds-(hours*3600 + minutes*60);
10     printf("Time is : \n");
11     printf("%d hours %d minutes %d seconds", hours, minutes, seconds);
12     return 0;
13 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 11_second_to_time.c
Enter total seconds:
19836
Time is :
5 hours 30 minutes 36 seconds
PS C:\VS CODE\cs_assignment_1> |
```

12. PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

```
cs_assignment_1 > C ASCII_CODE.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      char ch;
5      printf("Enter any character\n");
6      scanf("%c", &ch);
7      printf("ASCII value of %c is %d", ch, ch);
8      return 0;
9  }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc ASCII_CODE.c -o ASCII_CODE } ; if ($?) { .\ASCII_CODE }
Enter any character
k
ASCII value of k is 107
PS C:\VS CODE\cs_assignment_1> |
```

13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.

```
cs_assignment_1 > C even_or_odd.c > ...
1  #include <stdio.h>
2  int main()
3  {
4      int num;
5      printf("Enter any number\n");
6      scanf("%d", &num);
7      if (num % 2 == 0)
8      {
9          printf("%d is an even number", num);
10     }
11     else
12     printf("%d is an odd number", num);
13     return 0;
14 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) {
Enter any number
6
6 is an even number
PS C:\VS CODE\cs_assignment_1>
PS C:\VS CODE\cs_assignment_1> cd "c:\VS CODE\cs_assignment_
Enter any number
9
9 is an odd number
PS C:\VS CODE\cs_assignment_1> |
```

14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALL CASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

```
cs_assignment_1 > C 14_IDENTIFIER.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      char ch;
5      printf("Enter any character :");
6      scanf("%c", &ch);
7      //check character type using ASCII values
8      if (ch >= 'A' && ch <= 'Z')
9      {
10         printf("The character %c is a CAPITAL LETTER.\n", ch);
11     }
12     else if (ch >= 'a' && ch <= 'z')
13     {
14         printf("The character %c is a small case letter.\n", ch);
15     }
16     else if (ch >= '0' && ch <= '9')
17     {
18         printf("The character %c is a digit.\n", ch);
19     }
20     else
21     {
22         printf("The character %c is a special character.\n", ch);
23     }
24     return 0;
25 }
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 14_IDENTIFIER.c
Enter any character :l
The character l is a small case letter.
PS C:\VS CODE\cs_assignment_1> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc
Enter any character :*
The character * is a special character.
PS C:\VS CODE\cs_assignment_1> █
```

15. PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION.

cs_assignment_1 > C 15.c > ...

```
1  #include <stdio.h>
2  int main()
3  {
4      float s1, s2, s3, s4, s5, total, percentage;
5      printf("Enter the marks of 5 subjects: \n");
6      scanf("%f %f %f %f %f", &s1, &s2, &s3, &s4, &s5);
7      total=s1+s2+s3+s4+s5;
8      percentage=total/5;
9      printf("Total marks=%.2f\n", total);
10     printf("Total percentage=%.2f\n", percentage);
11     if(percentage>=70)
12     {
13         printf("GRADE A");
14     }
15     else if(percentage>=50 && percentage < 70)
16     {
17         printf("GRADE B");
18     }
19     else if(percentage>=35 && percentage<50)
20     {
21         printf("GRADE C");
22     }
23     else
24     {
25         printf("FAIL");
26     }
27     return 0;
28 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) {
Enter the marks of 5 subjects:
56
23
98
84
96
Total marks=357.00
Total percentage=71.40
GRADE A
```

16. PROGRAM TO READ THREE NOS. AND PRINT MAX

IOP_ASSIGNMENT > cs_assignment_1 > C MAX_NO_16.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int a, b, c;
5      printf("Enter values of a, b and c\n");
6      scanf("%d %d %d", &a, &b, &c);
7      if(a>b){
8          if(a>c){
9              printf("%d is the MAX number", a);
10         }
11         else{
12             printf("%d is the MAX number", c);
13         }
14     }
15     else{
16         if(b>c){
17             printf("%d is the MAX number", b);
18         }
19         else{
20             printf("%d is the MAX number", c);
21         }
22     }
23 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ;
Enter values of a, b and c
5
6
9
9 is the MAX number
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1>
```

17. PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR.(&&)

IOP_ASSIGNMENT > cs_assignment_1 > C MAX_no_17.c > main()

```
1  #include <stdio.h>
2  int main(){
3      int a, b, c;
4      printf("Enter value of a, b and c\n");
5      scanf("%d %d %d", &a, &b, &c);
6      if(a>b && a>c){
7          printf("%d is the MAX number", a);
8      }
9      else if(b>c && b>a){
10         printf("%d is the MAX number", b);
11     }
12     else{
13         printf("%d is the MAX number", c);
14     }
15 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\"
Enter value of a, b and c
9
3
8
9 is the MAX number
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1>
```

18. PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL (? :) OPERATOR.

```
IOP_ASSIGNMENT > cs_assignment_1 > C lConditional_operator_18.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      int a, b, c;
5      printf("Enter values of a, b and c\n");
6      scanf("%d %d %d", &a, &b, &c);
7      a>b ? (a>c ? printf("%d is MAX no. ", a) : printf("%d is MAX no.", c)) : (b>c ? printf("%d is the Max no. ", b) : printf("%d is the MAX no.", c));
8      return 0;
9  }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ; if ($?) { gcc lConditional_operator_18.c -o lConditional_operator_18 } ; if ($?) { .\lConditional_operator_18 }
Enter values of a, b and c
5
91
56
91 is the Max no.
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1>
```

19. PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDATIONAL OPERATOR.

```
IOP_ASSIGNMENT > cs_assignment_1 > C as1_19.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      char ch;
5      printf("Enter any character : ");
6      scanf("%c", &ch);
7      if(ch >= 'A' && ch <= 'Z'){
8          printf("character is not a small case letter");
9      }
10     else if(ch >='a' && ch<='z'){
11         printf("character is a small case letter");
12     }
13     else{
14         printf("Its not a letter");
15     }
16     return 0;
17 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ; if
Enter any character : l
character is a small case letter
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> cd "c:\VS CODE\IOP_ASS
Enter any character : K
character is not a small case letter
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> 
```

20. PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.

```
IOP_ASSIGNMENT > cs_assignment_1 > C 20_calculator.c > main()
1  #include<stdio.h>
2  int main(){
3      char sign;
4      float n1, n2;
5      printf("Enter n1 : ");
6      scanf("%f", &n1);
7      printf("Enter a operator (+, -, *, /) : ");
8      scanf(" %c", &sign);
9      printf("Enter n2 : ");
10     scanf("%f", &n2);
11     switch(sign){
12         case '+':
13             printf("sum is : %.2f", n1+n2);
14             break;
15         case '-':
16             printf("difference is : %.2f", n1-n2);
17             break;
18         case '*':
19             printf("multiplication is : %.2f", n1*n2);
20             break;
21         case '/':
22             printf("division is : %.2f", n1/n2);
23             break;
24     }
25 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ; i
Enter n1 : 6
Enter a operator (+, -, *, /) : *
Enter n2 : 5
multiplication is : 30.00
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> 
```

21. PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR) AND THIS PERFORM USING SWITCH-CASE. (CASE IN CHAR FORM).

IOP_ASSIGNMENT > cs_assignment_1 > C 21_aritmetic_op.c > ...

```
1  #include <stdio.h>
2
3  int main() {
4      double num1, num2;
5      char choice;
6      printf("Enter first number: ");
7      scanf("%lf", &num1);
8      printf("Enter second number: ");
9      scanf("%lf", &num2);
10
11     printf("\n--- Choose an Arithmetic Task ---\n");
12     printf("a. Addition (+)\n");
13     printf("s. Subtraction (-)\n");
14     printf("m. Multiplication (*)\n");
15     printf("d. Division (/)\n");
16     printf("Enter your choice (a, s, m, d): ");
17     scanf(" %c", &choice);
18     switch (choice) {
19         case 'a':
20             printf("Addition: %.2lf\n", num1 + num2);
21             break;
22         case 's':
23             printf("Subtraction: %.2lf\n", num1 - num2);
24             break;
25         case 'm':
26             printf("Multiplication: %.2lf\n", num1 * num2);
27             break;
28         case 'd':
29             if (num2 != 0) {
30                 printf("Division: %.2lf\n", num1 / num2);
31             } else {
32                 printf("Error! Cannot divide by zero.\n");
33             }
34             break;
35         default:
36             printf("Invalid choice! Please select a valid task option.\n");
37     }
38
39     return 0;
40 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ; if ($?) { gcc 21_aritmetic_op.
Enter first number: 6
Enter second number: 3

--- Choose an Arithmetic Task ---
a. Addition (+)
s. Subtraction (-)
m. Multiplication (*)
d. Division (/)
Enter your choice (a, s, m, d): a
Addition: 9.00
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> |
```

22. PROGRAM TO CALCULATE FACTORIAL OF A NO.

```
IOP_ASSIGNMENT > cs_assignment_1 > C 22_factorial.c > main()
1  #include <stdio.h>
2  int main(){
3      int number, i=1;
4      long int factorial;
5      factorial=1;
6      printf("Enter a number :");
7      scanf("%d", &number);
8      //M1//
9      //for(i=1; i<=number; i++){
10     //    factorial = i*factorial;
11     //}
12     while(i<=number){ //M2//
13         factorial=factorial*i;
14         i = i+1;
15     }
16     printf("%ld", factorial);
17     return 0;
18 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1"
Enter a number :6
720
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1>
```

23. PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE(<MATH.H>).

IOP_ASSIGNMENT > cs_assignment_1 > C 23_power.c > main()

```
1  #include<stdio.h>
2  int main(){
3      int power, number, i;
4      long int result;
5      printf("Enter number : ");
6      scanf("%d", &number);
7      printf("Enter power : ");
8      scanf("%d", &power);
9      result = 1;
10     /*for(i=1; i<=power; i++){
11         result = result * number;
12     }*/
13     i=1;
14     while(i<=power){
15         result = result * number;
16         i=i+1;
17     }
18     printf("The result is : %.2ld", result);
19 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1"
Enter number : 6
Enter power : 2
The result is : 36
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1"
Enter number : 
```

24. WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

$$Y(x,n) = \begin{cases} 1+x & \text{where } n=1 \\ 1+x/n & \text{where } n=2 \\ 1+x^n & \text{when } n=3 \\ 1+nx & \text{when } n>3 \text{ or } n<1 \end{cases}$$

IOP_ASSIGNMENT > cs_assignment_1 > C 24_value_of_Y.c > main()

```
1  #include <stdio.h>
2  #include <math.h>
3  int main(){
4      int x, n, a;
5      float Y;
6      printf("Enter value of x : ");
7      scanf("%d", &x);
8      printf("Enter value of n : ");
9      scanf("%d", &n);
10     if(n==1){
11         Y=x+1;
12         printf("Y = %f", Y);
13     }
14     else if(n==2){
15         Y = 1 + (x / n);
16         printf("Y = %f", Y);
17     }
18     else if(n==3){
19         Y = 1 + pow(x, n);
20         printf("Y = %f", Y);
21     }
22     else{
23         Y=1+(n*x);
24         printf("Y = %f", Y);
25     }
26 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ;
Enter value of x : 5
Enter value of n : 2
Y = 3.000000
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> |
```

25. PROGRAM TO PRINT TABLE OF ANY NO.

IOP_ASSIGNMENT > cs_assignment_1 > C 25_tableofanyno.c > main()

```
1  #include <stdio.h>
2  int main(){
3      int n, i;
4      printf("Enter the value of n :");
5      scanf("%d", &n);
6      for(i=1; i<=20; i++){
7          printf("%d*%d = %d\n", n, i, n*i);
8      }
9      return 0;
10 }
```

Problems Output Debug Console Terminal Ports

PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ; if (\$?) {

Enter the value of n :8

8*1 = 8

8*2 = 16

8*3 = 24

8*4 = 32

8*5 = 40

8*6 = 48

8*7 = 56

8*8 = 64

8*9 = 72

8*10 = 80

8*11 = 88

8*12 = 96

8*13 = 104

8*14 = 112

8*15 = 120

8*16 = 128

8*17 = 136

8*18 = 144

8*19 = 152

8*20 = 160

PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> █

26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N

IOP_ASSIGNMENT > cs_assignment_1 > C 26_oddevensumtoN.c > main()

```
1  #include <stdio.h>
2  int main(){
3      int N, i, even_sum, odd_sum;
4      even_sum = 0;
5      odd_sum = 0;
6      printf("Enter a positive integer N: ");
7      scanf("%d", &N);
8      for(i=1; i<=N; i++){
9          if(i%2==0){
10             even_sum=even_sum+i;
11         }
12         else{
13             odd_sum=odd_sum+i;
14         }
15     }
16     printf("Sum of even numbers from 1 to %d is : %d\n", N, even_sum);
17     printf("Sum of odd numbers from 1 to %d is : %d\n", N, odd_sum);
18     return 0;
19 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ; if ($?) { gcc
Enter a positive integer N: 8
Sum of even numbers from 1 to 8 is : 20
Sum of odd numbers from 1 to 8 is : 16
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1> 
```